

The CAPM is not wrong because beta fails.

It is wrong for a much more
interesting reason.

Where asset pricing models actually come from, why finding betas proves almost nothing, and the one number that damages the consumption model without any estimation at all.

Ends with three objections to my own argument, and a question I would like to be argued with on.

Everything is one equation

01 — THE FRAMEWORK

Mean–variance analysis, the CAPM, Fama–French, the consumption model. They look like competing theories. They are not.

$$p = \mathbb{E}[mx]$$

Price today = expected payoff, weighted by a *stochastic discount factor* m . Every model in that list is this equation with a different guess at m .

And m is not an assumption you have to justify. A strictly positive m exists **if and only if** there is no arbitrage. That is the whole requirement. No utility function, no equilibrium, no rationality.

If you accept that free money doesn't sit on the pavement, you have already accepted the pricing kernel.

Why risk premia are about covariance

02 — THE ONLY RESULT YOU NEED

Expand $1 = \mathbb{E}[mR]$ using the definition of covariance and you get:

$$\mathbb{E}[R^i] - R^f = -R^f \text{Cov}(m, R^i)$$

Read it carefully. An asset earns a premium **only** if it pays badly when m is high — when money is scarce and valuable.

Variance appears nowhere. An asset with enormous volatility that is uncorrelated with m earns exactly the risk-free rate.

This is the CAPM's intuition about idiosyncratic risk, except it holds with no assumptions about normality, no quadratic utility, and no market portfolio.

The CAPM's famous insight is not the CAPM's. It falls out of no-arbitrage alone.

Beta pricing is a rewriting, not a theory

03 — THE PART PEOPLE GET WRONG

Assume the kernel is linear in some factors, $m = a + b'f$. Then

$$\mathbb{E}[R^i] = R^f + \beta_i' \lambda$$

and — this is the important half — the converse is also true. Any expected-return–beta relation can be rewritten as a linear factor model.

So: finding that expected returns line up with betas is **not** evidence for a theory. Give me enough freedom over f and I will always produce that relation. The extreme case: take the factor to be the mean–variance efficient portfolio, which always exists.

The content of a model is which factor, chosen in advance, and why it should be priced. Not the fit.

The CAPM's claim is specific: the factor is the return on aggregate wealth. Its price of admission is quadratic utility or normal returns, plus consumption equal to wealth.

One number, no free parameters

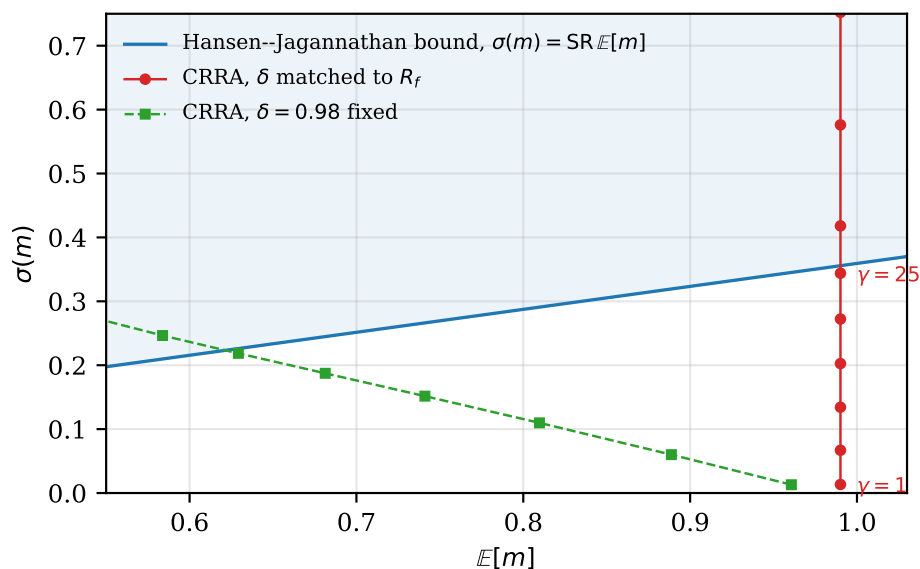
04 — THE TEST THAT NEEDS NO ESTIMATION

For any excess return, $\mathbb{E}[mR^e] = 0$. Take covariances, use $|\text{corr}| \leq 1$:

$$\frac{\sigma(m)}{\mathbb{E}[m]} \geq \frac{|\mathbb{E}[R^e]|}{\sigma(R^e)} = \text{Sharpe ratio}$$

The Hansen–Jagannathan bound. No parameters to fit, no regression to run. The market offers a Sharpe ratio around 0.36, so *any* valid kernel must be at least that volatile relative to its mean.

Now try the textbook kernel, $m = \delta C_1/C_0$ raised to $-\gamma$. Its coefficient of variation is roughly $\gamma\sigma_c$, and aggregate consumption growth has $\sigma_c \approx 1.35\%$.



To clear the bound you need $\gamma = 25.8$. Impose the observed consumption–return correlation of 0.2 and you need 133.

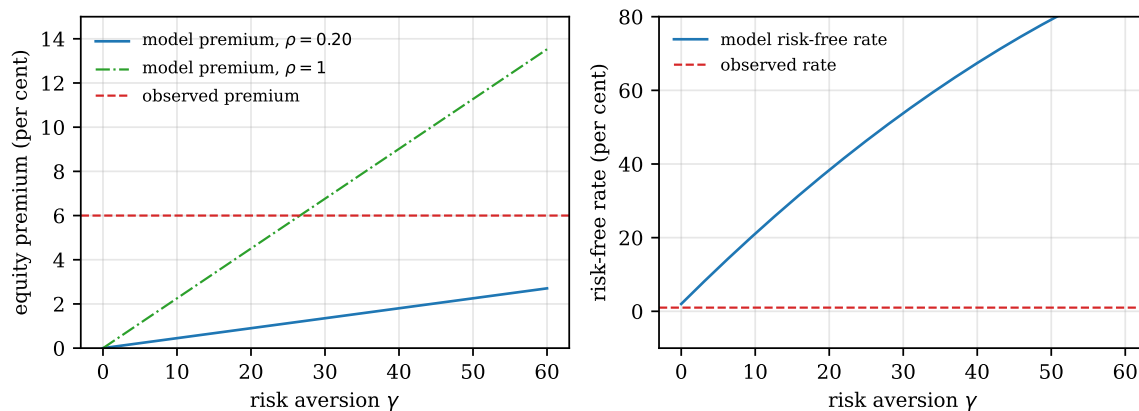
And then the second bill arrives

05 — WHY RAISING γ DOESN'T RESCUE IT

Fine, say risk aversion is huge. But the same parameter also sets the risk-free rate:

$$r^f = \rho + \gamma g - \frac{1}{2} \gamma^2 \sigma_c^2$$

A very risk-averse investor facing consumption that grows at 2% a year desperately wants to borrow. In equilibrium the rate must rise to stop him.



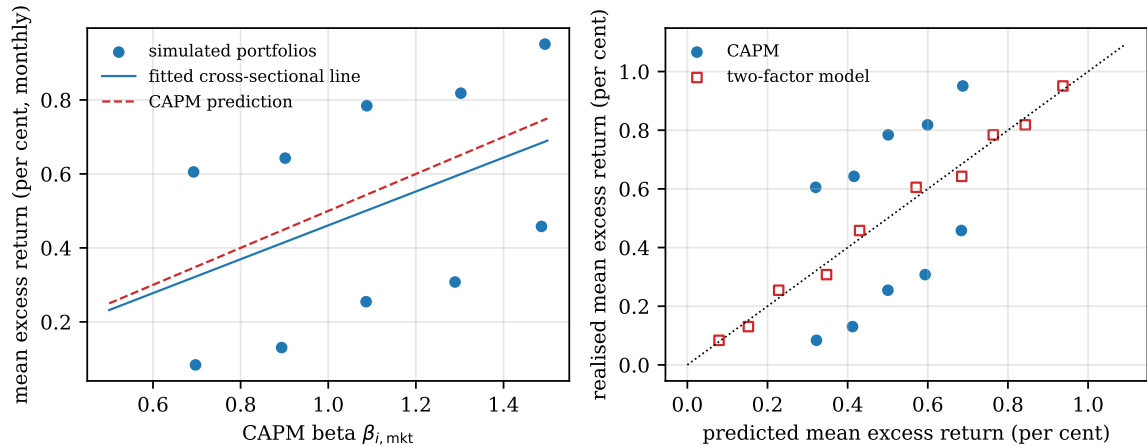
At $\gamma = 27.8$ the model predicts a real risk-free rate of **50%**. At $\gamma = 133$, over **100%**. Observed: about 1%.

These are not two puzzles. They are one fact: aggregate consumption is far too smooth to be a plausible source of marginal-utility volatility.

What a missing factor looks like from the inside

06 — A SIMULATION, AND AN UNCOMFORTABLE POINT

I built an economy where the true kernel has *two* priced factors, then priced ten portfolios with a single-factor model. 600 months, orthogonal design.



Result: every beta recovered accurately. Every asset mispriced. The alphas line up perfectly with loadings on the omitted factor, t -statistics of 3.5 to 4.9, and the joint test rejects at $p = 1.1 \times 10^{-4}$. Add the second factor and the alphas vanish.

Estimating exposures well and pricing well are different achievements. Only the second tests a model.

The uncomfortable part: in real data you see the alphas without knowing the true kernel. A missing risk factor, a missing conditioning variable, and a data-mined coincidence all look identical in that chart.

Three objections to all of this

07 — AND WHAT I WOULD SAY BACK

1. “Your correlation of 0.2 is too low. Consumption data is time-averaged and measured with error.”

Correct, and it is the strongest attack on the calculation. Correcting for time aggregation raises the effective correlation and lowers the required γ — but only towards the Hansen–Jagannathan value of about 26, which is still an order of magnitude above anything defensible. And it does nothing at all to the risk-free rate, which involves no correlation.

2. “The bound binds only for a kernel perfectly correlated with the market, so it is not a serious test.”

That is backwards. Perfect correlation is the *most generous* case. Any realistic kernel is imperfectly correlated and therefore needs *more* volatility, not less. Failing the most permissive test available is worse than failing a specific model, not better.

3. “The CAPM is a conditional model. Unconditional tests reject it unfairly.”

Genuinely right, and the point I find hardest to dismiss. A model can hold conditionally and fail unconditionally when betas and premia move over time. But it cuts both ways: it also makes the model far harder to *use*, since you must specify the conditioning information, and it does not touch the consumption puzzles, which are statements about unconditional first and second moments.

If any of these is wrong, I would rather find out in the comments than in an interview.

So where does that leave us?

08 — THE OPEN QUESTION

The framework survives all of this. $p = \mathbb{E}[mx]$ follows from no-arbitrage and a first-order condition; the Hansen–Jagannathan bound is a valid diagnostic against anything. What fails is the specification of m .

Every serious response tries to make marginal utility volatile *without* making consumption volatile:

- **Habit** (Campbell–Cochrane) — risk aversion moves with the distance from a slow-moving habit.
- **Long-run risk** (Bansal–Yaron) — a small persistent component in growth makes today’s shock revise a lifetime of expectations.
- **Epstein–Zin** — break the mechanical tie between risk aversion and intertemporal substitution, so the premium and the risk-free rate can be set separately.

Each works for the calibration it was designed for. None has won.

My question, and I would like to be argued with:

Is the consumption-based programme worth continuing, or is the honest position that we have descriptive factor models that work and no accepted story for why those factors are risky?

Full write-up with proofs, calibration and Python code in the comments.

Corrections welcome — particularly on the calibration choices.

Where to read this properly

09 — REFERENCES

The framework

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The three responses

Campbell and Cochrane, *J. Political Economy* 107 (1999) — habit.

Bansal and Yaron, *J. Finance* 59 (2004) — long-run risk.

Epstein and Zin, *Econometrica* 57 (1989) — recursive preferences.

Calibration inputs are documented magnitudes from Mehra–Prescott and Campbell, not estimates of my own. The simulated economy of slide 7 is synthetic and labelled as such.